Generalized Multimodal Agents

CAP6411

General Multimodal AI Agents

- Multimodal Understanding
  - to understand multimodal input from various domains (both digital and physical) not only semantically, but also spatially and temporally.
- Multimodal Action Prediction
  - to translate a task into an action sequence and execute the actions with precision in various environments.

Tasks

- Input: image/video + textual description of tasks
- UI navigation in 2D screenshots.
  - The task could be “book a hotel” and the output should include both semantic type of action (e.g., “type”, “click”, etc.) and the location (x,y) or box (x,y,w,h) where the action is applied.
- Robot manipulation in the 3D world.
  - For a task like “close the drawer”, the output consists of 6-DoF displacements (x,y,z,yaw,pitch,roll) of the end effector and an extra dimension to indicate gripper openness.
- Multimodal understanding tasks.
  - For a regular task like VQA, the problem is reduced to a multimodal understanding that generates a textual description and/or location of objects.

User: What is they doing?
Magma: They are camping in a forest and hiking on a rocky mountain.
User: On what date did the individual leave a place that Simon thought was very important to him?
Magma: The date is June 9, 2021.
Temporal
User: The figure represents a 5x5 grid containing various animals, where each 1x1 square is considered a block and each block contains an animal from ['cat', 'dog', 'elephant', 'giraffe', 'rabbit']. What is the animal of the block located at the first row, second column of the grid?
Magma: The animal is a cat.
Spatial
User: What is unusual thing about this image?
Magma: The unusual aspect of this image is that a man is ironing clothes on the back of a yellow taxi. ....
User: How to increase the sound?
Magma: Tap on the "Volume" icon. Then, slide the volume up to increase the volume.
Multimodal Understanding
Task: Push the chip bag to left
Task: Put the can to left
Task: Peel the potato
Task: Stir the food in the pot
Planning
Task: Plan a visit to the museum
Task: Lock Screen
Task: View my Account
Multimodal Action Grounding
User: Close the middle drawer
Magma: Trajectory Rollout on SimplerEnv Google Robot
User: Pick up the mushroom to the pot
Magma: Trajectory Rollout on Widow-X Real Robot
User: Find baby shoes under 20 dollars and rated 5 stars.
Magma: Trajectory on Mind2Web
Task: Install app "Instagram"
Magma: Trajectory on AITW
Multimodal Agentic Taks
Physical Environment
Digital Environment

Training Dataset

- A large-scale pretraining dataset with a total size of 39M diverse samples, coming from not only VL datasets, but also UI, robotics data and human instructional videos, ***all autolabeled using SoM and ToM.***

Set-of-Mark (SoM)

- Image segmentation
- DOM Tree for web
- Given candidate bboxes:
  - Draw boxes
  - Box pooling (NMS)
  - Textbox size, coordinates, label
- After training:
  - Omniparser provides the candidates
  - Run model pretrained on SoM

Trace-of-Mark (ToM)

- Utilize point tracker CoTracker
  - Generate moving trajectories at pixel level
- Start from timestep t
- Given video of length l,
  - CoTracker extract (l-t) grid of equally spaced points
  - Run homography to mitigate noise
  - Classify into fg and bg traces based on average motion
  - Kmean (fg and bg) to select one or more points from each cluster as the final traces